

Supporting Information for

Do LLMs write like humans? Variation in grammatical and rhetorical styles

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Supporting Information Text

Construction of Corpora. For this study, we created two parallel corpora. The construction of each corpus was accomplished by:

1. assembling 12,000 human-authored texts in English representing a range of registers (or types);
2. selecting roughly the first 1,000 words of each text (where that span does not cross a section or chapter boundary);
3. splitting each string into approximately 500-word chunks (with the split-point being a sentence boundary as determined by a spaCy dependency parse);
4. passing the first chunk to six different LLMs and prompting the models to generate 500 more words of text in the same style, similar to the method used by (1);
5. discarding all texts that contain a non-response or a nonsense response (like a string of punctuation) from any of the six LLMs.

The resulting corpora, then, contain the first chunk of human-authored text (the "germinal" text used for the LLM prompt) and seven variations of the second chunk: the original (human-authored) continuation of the first chunk, and continuations generated by six different LLMs. This workflow is illustrated in Figure TODO of the main text. The figure also highlights the alignment of the second, parallel chunks, which enable comparisons between the writing of humans and LLMs, as well as comparisons among the writing produced by the LLMs, themselves.

The first of our parallel corpora uses the Corpus of Contemporary American English (COCA) as the source of its human-authored text and is called the COCA AI Parallel (CAP) corpus. As noted above, we wanted data representative of a broad range of sources and registers. COCA is a monitor corpus with over 1 billion words and balanced among eight registers: spoken, fiction, magazines, newspapers, academic journals, blogs, other web pages, and TV/movie subtitles (2). For our dataset, we filtered for texts containing a minimum of approximately 1,250 words, then randomly sampled 1,500 documents from each register to create a corpus of 12,000 texts.

Though its breadth and representativeness make COCA valuable for this task, it does have a key limitation: to allow distribution of otherwise copyrighted material, COCA alters the text by removing 10 words every 200 words.[*] This produces gaps and incomplete sentences in each 500-word chunk that might affect sentence parsing and feature extraction. For this reason, we constructed a second corpus, which is meant to approximate COCA's composition.

The resulting dataset, the Human AI Parallel English (HAP-E) corpus, is drawn from six (as opposed to COCA's eight) text-types:[†]

- **Academic**: samples were drawn from a 40,000+ document corpus of open-access academic articles published by Elsevier.
- **News**: samples were drawn from a 100,000+ document corpus of news articles published online by U.S.-based news organizations.
- **Fiction**: samples were drawn from novels and short stories in the public domain and available on Project Gutenberg.
- **Spoken**: samples were drawn from a corpus of 100,000 podcast transcriptions. (Note that COCA's spoken text-type comes from transcriptions of unscripted news shows.)
- **Blogs**: samples were drawn from a corpus of 681,288 posts from blogger.com.
- **Television and Movie Scripts**: samples were drawn from two different corpora of scripts, some of which were converted using OCR.

The sampling and processing of the HAP-E corpus followed the same five steps described at the beginning of this section. The word counts of the HAP-E and CAP corpora can be found in the Appendix, Tables S2 and S3.

[*] See https://www.corpusdata.org/limitations.asp for details on the procedure.

[†] The data can be accessed from https://huggingface.co/datasets/browndw/human-ai-parallel-corpus.

LLM text generation. We used multiple versions of GPT-4o (3, 4) and Meta Llama 3 (5) to generate the parallel LLM texts, shown in Table S1. We chose these LLMs because they are popular, available within a reasonable budget, and generally considered to perform very well. Meta Llama 3 is available in both base models, which provide text completion, and instruction-tuned variants that use additional human feedback to optimize the models to follow instructions and answer questions; recent GPT variants are only available in instruction-tuned form. Llama 3's availability in both forms allows us to observe the effects of instruction tuning.

Instruction-tuned LLMs were given the following prompt, followed by the first chunk of human text:

> In the same style, tone, and diction of the following text, complete the next 500 words, generate exactly 500 words, and note that the text does not necessarily end after the generated words:

Base Llama 3 variants without instruction tuning were simply given the first chunk of text. In each case, the LLMs generated roughly 500 words of additional output, continuing where the first chunk finished. This could be compared directly to the second chunk of human text. GPT-4o text was generated using OpenAI's Batch API using its default settings; Llama 3 text was generated on a local GPU cluster using the vLLM package for Python, version 0.5.5 (6).

As some LLMs refused to complete some prompts, or returned short, unusable answers, we removed responses shorter than 100 words, producing 8,290 HAP-E texts and 9,615 COCA texts with valid responses from all LLMs. Word counts in HAP-E and CAP by text-type are shown in Tables S2 and S3.

We designed this procedure for creating parallel data because our aims are focused more on description than they are on detection. Compared to other methods that only prompt LLMs with a title or topic (e.g. 7), this approach gives the LLMs roughly 500 words of context so they can adapt their output and its style to the human text.

Feature extraction. To extract meaningful features from our corpus for training our classifiers, we used Douglas Biber's tagset of 66 linguistic categories, gleaned from the many works Biber has authored and co-authored (8–10). Biber's tagset includes indices of lexical complexity (e.g., mean word length), but many more are raw linguistic features ranging from the lexical (e.g., thinking verbs) to the grammatical (e.g., adjectives before a noun). For example, features include:

- Nominalizations (nouns formed by adding suffixes to verbs or adjectives, such as "justification", "development", "robustness")
- Agentless passive voice (such as "The model was fitted")
- Hedging phrases (such as "at about", "something like", "almost")
- Phrasal coordination (such as using "and" to coordinate between noun phrases)
- Clausal coordination (such as using "and" at the beginning of clauses).

A full list of features is provided in Table S4. Combined, these variables give a stylistic sense of how these texts read. These stylistic profiles, in turn, can enrich our understandings of the affordances and limitations of LLMs as they are currently developed and trained.

To tag the texts, we used the pseudobibeR package for R, which counts Biber features present in dependency-parsed texts.[‡] pseudobibeR has proven effective at reproducing results similar to those of other studies that use Biber's tagset (11, 12). Once tagged, we counted the rate of occurrence of each feature per 1,000 words of text. These rates served as the features for our analysis.

Corpus linguists studying language use and variation have regularly turned to counts of morphosyntactic and lexical features as the basis for their quantitative analysis, an approach which dominates the field (8, 13–15). Biber's tagset offers a compelling approach to studies of linguistic variation: because linguistic features both routinely co-occur and are sensitive to context (11), analysis of feature groupings that vary across sources lends insight into the salient distinctions between the stylistic profiles of human and LLM texts. We chose this tagset to represent this perspective, but also because it has been validated by a considerable amount of scholarship (16, 17). Of course, other foundational corpus-based studies have used different feature sets or "bag-of-words" methods, generating word frequency lists to investigate keywords, or for instance, markers of stance in academic writing (18, 19), Instead, Biber's morphosyntactic tagset presents a more robust method for analyzing register and genre, capturing a wide range of linguistic functions.

[‡] Available on the Comprehensive R Archive Network (CRAN) and at https://github.com/browndw/pseudobibeR

Statistical analysis. To compare the rate of feature use between human and LLM texts, we compared each LLM's rate of use with the human chunk 2 usage. Table S5 shows the usage (per 1,000 tokens) of each feature in human text, and the LLM usage scaled relative to the human usage of each feature. To scale these differences relative to the typical variation in usage across texts, we calculated the difference between the LLM and human rates for each individual text, then calculated the Cohen's d effect size by dividing the mean difference by the standard deviation of the differences. These effect sizes are shown in Table S6. Typically, $d \leq 0.5$ is considered a small effect, $0.5 < d \leq 0.8$ is considered medium, $0.8 < d \leq 1.3$ is large, and $d > 1.3$ is very large (20). Differences were tested for significance using the paired Wilcoxon signed-rank test.

Classification. To evaluate whether the features described above are sufficient to distinguish between human- and LLM-written text, we randomly split the HAP-E documents into a training set (75%) and a test set (25%), stratifying by document category.

First, we fit a random forest to classify the source of each of the texts using the Biber features. This is a seven-class classification task (human and six LLMs). This random forest identified the features most important for distinguishing between sources; these top features are used for visualizations of feature variations by source. This was fit using the `ranger` package (21) with its default settings.

Next, we trained a random forest for each pairwise human vs. LLM training task: human vs. GPT-4o, human vs. Llama 3 8B, and so on. For comparison, we also fit lasso-penalized logistic regression models using the same set of Biber features. We used the performance of the lasso models to benchmark and provide context to the random forest models. Since we only account for linear additive contributions of the Biber features in the lasso models, any noticeable improvements in performance of the random forests are due to the flexibility of tree-based models in capturing the presence of non-linear interactions. All lasso logistic regression models were fit using the `glmnet` package with the one-standard-error regularization penalty selected with 10-fold cross-validation (22). Accuracies of the random forest and lasso models on the test set are shown in Tables S9 and S10.

Finally, we used these models to classify text from the M4 parallel corpus (7). We focus on its arXiv GPT-3.5 dataset, which includes both real abstracts of preprints posted on arXiv and text generated by GPT-3.5 when prompted to write an abstract based on the preprint's title. This provides an external test set, generated on a different task with a different model, to evaluate the generalizability of our classifiers. The accuracies on this test set are shown in Tables S11 and S12.

Detection has been a particular emphasis since the release of ChatGPT and similar LLMs for understandable reasons (23, 24). While our work involves similar classification tasks, we use classification not as an end in and of itself, but rather as method for helping us understand how LLMs write: how the text that they produce compares to the writing humans produce and how the text produced by one LLM compares to that produced by another.

Model	Version	Context	Instruct?
GPT-4o Mini	2024-07-18	128K	Yes
GPT-4o	2024-08-06	128K	Yes
Llama 3	8B	8K	No
Llama 3	8B-Instruct	8K	Yes
Llama 3	70B	8K	No
Llama 3	70B-Instruct	8K	Yes

Table S1. Large language models used to generate text completions.

Table S2. Word counts in the HAP-E corpus, by source

Author	acad (n = 1227)	blog (n = 1526)	fic (n = 1395)	news (n = 1322)	spok (n = 1721)	tvm (n = 1099)	Total
Human							
Chunk 1	602,878	748,901	674,437	644,896	833,699	598,157	4,102,968
Chunk 2	602,683	749,052	675,595	646,030	832,797	591,704	4,097,861
ChatGPT							
GPT-4o	633,884	826,590	759,461	708,375	969,210	606,059	4,503,579
GPT-4o Mini	690,828	916,433	837,527	771,324	1,036,125	678,429	4,930,666
Llama Base							
Llama 3 70B	510,942	731,889	651,557	572,423	1,098,698	502,012	4,067,521
Llama 3 8B	544,845	735,068	740,610	510,671	1,083,953	525,037	4,140,184
Llama Instruction-Tuned							
Llama 3 70B Instruct	580,579	735,501	641,982	627,299	934,911	453,559	3,973,831
Llama 3 8B Instruct	5550,463	679,884	599,000	568,703	836,531	448,874	3,683,455
Total	4,717,102	6,123,318	5,580,169	5,049,721	7,625,924	4,403,831	33,500,065

Table S3. Word counts in the CAP corpus, by source

Author	acad (n = 1221)	blog (n = 1071)	fic (n = 1372)	mag (n = 1293)	news (n = 1188)	spok (n = 1334)	tvm (n = 1006)	web (n = 1130)	Total
Human									
Chunk 1	594,011	521,050	669,470	629,726	579,755	651,956	497,733	549,341	4,693,042
Chunk 2	593,875	521,493	671,400	630,126	580,208	651,013	497,900	549,736	4,695,751
ChatGPT									
GPT-4o	646,061	578,298	754,715	696,988	640,690	734,320	562,464	609,670	5,223,206
GPT-4o Mini	710,560	631,028	830,929	759,785	697,919	793,067	608,491	664,693	5,696,472
Llama Base									
Llama 3 70B	634,881	517,613	739,502	681,171	583,279	747,996	464,058	570,922	4,939,422
Llama 3 8B	625,065	527,628	729,348	686,137	553,703	736,093	462,948	550,569	4,871,491
Llama Instruction-Tuned									
Llama 3 70B Instruct	607,677	525,063	637,822	632,915	561,171	694,292	406,103	549,999	4,615,042
Llama 3 8B Instruct	554,642	480,391	597,560	572,409	514,765	624,254	409,802	508,688	4,262,511
Total	4,966,772	4,302,564	5,630,746	5,289,257	4,711,490	5,632,991	3,909,499	4,553,618	38,996,937

Table S4. Lexical, grammatical, and rhetorical features detected by pseudobibeR, adapted from (8, Appendix II).

Feature	Description	Examples
Past tense	Verbs in past tense	I *ran* far.
Perfect aspect	Verbs in perfect aspect	I *have* written this sentence.
Present tense	Verbs in present tense.	the duck *walks* quickly
Place adverbials	Adverbs and adverbial phrases describing place	above, beside, outdoors
First-person pronouns		I, we, our
Second-person pronouns		you, your
Third-person pronouns	(excluding *it*)	he, she, they, their
Pronoun *it*		it, its, itself
Demonstrative pronouns	Pronouns replacing nouns	*That* is an example sentence.
Indefinite pronouns		anybody, nothing, someone
Pro-verb 'do'		
'Wh-' questions	Direct who, what, when, where, and why questions	*When* are you leaving?
Nominalizations	Nouns ending in *-tion*, *-ment*, etc.	justification, development, robustness
Gerunds	Participial forms functioning as nouns	
Other nouns	All other nouns	
Agentless passives	Agentless passive voice	The model *was fitted*
'By-' passives	Passive voice with agent	The task *was done by Steve*
'Be' as main verb	Use of "be" forms as main verb	
Existential 'there'	"There" used to assert something exists	*There* is a feature in this sentence
'That' verb complements		I said *that he went*
'That' adjective complements		I'm glad *that you like it*
'Wh-' clauses	Clauses beginning with 'wh-' words (*who*, *what*, *when*, ...)	I believed *what he told me*
Infinitives	Uninflected verb preceded by *to*	she tried *to explain*
Present participial clauses	Adverbial clauses used as present participles	*Stuffing his mouth with cookies,* Joe ran out the door
Past participial clauses	Adverbial clauses used as past participles	*Built in a single week,* the house would stand for fifty years
Past participial postnominal	Reduced relative past participial clauses	The solution *produced by this process*
Present participial postnominal	Reduced relative present participial clauses	The event *causing this decline*
'That' clauses as subject	'That' relative clauses in subject position	the dog *that bit me*
'That' clauses as object	'That' relative clauses in object position	the dog *that I saw*
'Wh-' relatives as subject	'Wh-' relatives in subject position	the man *who likes popcorn*
'Wh-' relatives as object	'Wh-' relatives in object position	the man *who Sally likes*
Pied-piping relative clauses	Relative clauses moved in sentence by 'wh-' questions	the manner *in which he was told*

Table S4. Lexical, grammatical, and rhetorical features (cont'd)

Feature	Description	Examples
Sentence relatives		Bob likes fried mangoes, *which is the most disgusting thing I've ever heard of*
Because	Causative adverbial subordinator	because
Though	Concessive adverbial subordinators	although, though
If, unless	Conditional adverbial subordinators	if, unless
Other adverbial subordinators		since, while, whereas
Prepositional phrases		
Attributive adjectives		the *big* horse
Predicative adjectives		the horse is *big*
Adverbs	Total adverbs	
Type-token ratio	Type-token ratio, including punctuation	
Mean word length	Average word length in characters, excluding punctuation	
Conjuncts		consequently, furthermore, however
Downtoners		barely, nearly, slightly
Hedges		at about, something like, almost
Amplifiers		absolutely, extremely, perfectly
Emphatics		a lot, for sure, really
Discourse particles		sentence-initial *well*, *now*, *anyway*
Demonstratives	"That", "this", "these", or "those" used as determiners	*That* is the feature
Possibility modals		can, may, might, could
Necessity modals		ought, should, must
Predictive modals		will, would, shall
Public verbs	Verbs indicating speaking or announcing	assert, declare, mention, predict, swear
Private verbs	Verbs expressing a personal intellectual state	assume, believe, doubt, presume, understand
Suasive verbs	Verbs indicating persuasion	command, insist, propose
'Seem' and 'appear'	Used as verbs	
Contractions		can't, won't
'That' deletion	'That' omitted as subordinator	I think *[that]* he went
Stranded prepositions	Preposition at the end of sentence or clause rather than before its object	she is the candidate that I was thinking *of*
Split infinitives	Infinitive with adverb between *to* and the verb	he wants *to convincingly prove* that...
Split auxiliaries		they *were apparently shown* to...
Phrasal co-ordination	Pairs of nouns, verbs, adjectives, or adverbs connected by a coordinating conjunction	The *nouns and verbs* are coordinated
Clausal co-ordination	Sentence clauses coordinated by a conjunction	The sentence was long, *but* I read it anyway
Synthetic negation		no answer is good enough
Analytic negation		that isn't good enough

Table S5. Biber features distinguishing human- and LLM-written text in HAP-E
Rate per 1,000 tokens; LLM rates relative to human Chunk 2. Blue indicates higher-than-human usage, red indicates lower. Ordered by feature importance for the seven-class classification task.

	Human		GPT		Llama 3 Instruct		Llama 3 Base		
Feature	Chunk 1	Chunk 2	GPT-4o Mini	GPT-4o	8B	70B	8B	70B	Importance
Present participial clauses	1.7	1.7	481%	527%	224%	261%	94%	102%	1,715.1
'That' clauses as subject	2.1	2.1	331%	263%	180%	173%	64%	68%	1,387.2
Mean word length	4.5	4.4	114%	116%	101%	103%	99%	100%	1,070.4
Adverbs	67.9	71.8	86%	82%	73%	75%	102%	102%	916.0
Nominalizations	14.6	14.6	209%	214%	145%	151%	88%	91%	808.9
Phrasal co-ordination	6.7	6.1	144%	194%	187%	170%	92%	97%	776.0
Clausal co-ordination	11.2	12.4	63%	59%	141%	127%	120%	116%	727.3
Other nouns	254.9	240.6	97%	103%	91%	95%	91%	94%	725.5
Downtoners	1.9	1.9	155%	118%	60%	57%	68%	73%	712.8
Attributive adjectives	46.7	43.8	140%	150%	100%	104%	79%	83%	677.7
Demonstratives	6.2	6.5	137%	133%	77%	80%	75%	80%	675.9
Agentless passives	7.7	7.8	51%	53%	96%	89%	101%	98%	664.4
Past participial postnominal	1.6	1.5	257%	235%	75%	75%	129%	131%	633.7
Infinitives	15.9	16.5	87%	83%	140%	132%	120%	113%	614.9
Prepositional phrases	101.0	98.0	118%	118%	100%	100%	87%	90%	581.2
Emphatics	8.6	9.2	75%	68%	97%	98%	76%	75%	545.9
Existential 'there'	1.9	2.1	59%	71%	42%	42%	108%	109%	527.3
Possibility modals	5.3	5.7	144%	104%	111%	116%	99%	99%	521.1
Past tense	37.5	41.9	77%	83%	91%	83%	115%	111%	519.0
'That' verb complements	2.1	2.5	70%	55%	144%	114%	159%	147%	503.4
Contractions	16.9	18.1	63%	60%	141%	139%	142%	129%	497.7
Other adverbial subordinators	5.9	6.2	136%	118%	114%	107%	82%	89%	496.6
'Be' as main verb	29.0	30.0	61%	63%	107%	100%	108%	101%	494.3
Public verbs	5.8	6.8	53%	63%	65%	67%	119%	112%	466.1
Present participial postnominal	1.4	1.3	293%	243%	125%	124%	101%	113%	451.0
Predicative adjectives	5.8	6.2	84%	90%	140%	131%	169%	154%	450.8
Suasive verbs	2.9	3.2	98%	116%	89%	90%	141%	137%	449.6
Analytic negation	8.5	9.7	73%	61%	80%	78%	113%	107%	439.6
Third-person pronouns	26.2	29.8	91%	91%	108%	104%	123%	120%	436.8
Private verbs	16.0	18.1	91%	85%	122%	113%	128%	126%	435.9
Perfect aspect	7.5	7.2	62%	60%	121%	111%	96%	92%	434.2
Pro-verb 'do'	2.9	3.2	25%	26%	60%	59%	115%	109%	430.8

Table S5. Biber features distinguishing human- and LLM-written text in HAP-E (cont'd)

	Human		GPT		Llama 3 Instruct		Llama 3 Base		
Feature	Chunk 1	Chunk 2	GPT-4o Mini	GPT-4o	8B	70B	8B	70B	Importance
Present tense	57.2	58.3	82%	76%	99%	100%	119%	116%	418.9
Amplifiers	2.0	2.1	63%	46%	40%	45%	85%	85%	412.6
Pronoun 'it'	12.5	13.2	88%	92%	105%	105%	108%	105%	412.0
Gerunds	3.0	3.0	156%	152%	119%	124%	83%	86%	402.9
First-person pronouns	34.0	35.3	81%	62%	111%	108%	136%	127%	393.4
Because	1.2	1.5	19%	20%	38%	38%	121%	105%	387.9
Split auxiliaries	3.4	3.4	91%	77%	126%	118%	98%	92%	379.4
Time adverbials	3.7	3.7	77%	79%	68%	70%	115%	119%	366.0
Though, although	0.5	0.5	65%	129%	21%	28%	83%	90%	361.2
Place adverbials	3.4	3.4	146%	144%	101%	99%	118%	128%	353.9
Demonstrative pronouns	5.5	6.1	55%	50%	71%	76%	99%	97%	350.8
Predictive modals	5.1	5.6	72%	57%	95%	106%	118%	111%	345.4
Indefinite pronouns	3.2	3.3	73%	77%	84%	83%	156%	149%	323.5
by-passives	0.9	0.9	56%	63%	118%	107%	113%	103%	314.9
Synthetic negation	1.2	1.3	36%	51%	36%	36%	93%	91%	311.8
'Wh-' questions	1.3	1.4	89%	56%	141%	137%	140%	132%	306.1
If, unless	2.2	2.6	78%	60%	58%	62%	117%	111%	305.1
'Seem' and 'appear'	0.7	0.7	131%	179%	140%	128%	99%	105%	290.3
Necessity modals	1.1	1.3	105%	78%	51%	54%	143%	137%	286.7
Conjuncts	2.1	2.3	110%	96%	97%	116%	99%	104%	281.2
Second-person pronouns	14.9	15.7	63%	52%	77%	81%	118%	110%	277.3
Stranded prepositions	0.8	0.9	66%	66%	111%	114%	87%	81%	266.5
'Wh-' relatives as subject	2.3	2.1	70%	66%	109%	102%	106%	103%	259.1
Sentence relatives	1.0	1.0	50%	51%	104%	117%	101%	92%	251.7
'Wh-' clauses	1.7	1.9	75%	66%	90%	78%	126%	122%	247.1
Discourse particles	1.1	1.0	60%	60%	89%	91%	146%	140%	181.9
Hedges	1.1	1.3	50%	63%	62%	67%	89%	92%	181.1
Past participial clauses	0.3	0.3	273%	307%	158%	150%	59%	71%	171.7
'That' deletion	0.7	0.8	75%	66%	75%	73%	134%	151%	169.8
Pied-piping relative clauses	0.6	0.6	59%	56%	38%	43%	50%	54%	164.5
'That' clauses as object	0.6	0.6	65%	56%	66%	60%	93%	90%	118.3
'That' adjective complements	0.3	0.4	80%	51%	121%	106%	148%	139%	107.6
Split infinitives	0.2	0.3	79%	94%	110%	104%	44%	49%	94.9
'Wh-' relatives as object	0.3	0.3	20%	13%	8%	13%	61%	62%	62.5

Table S5. Biber features distinguishing human- and LLM-written text in HAP-E (cont'd)

	Human		GPT		Llama 3 Instruct		Llama 3 Base		
Feature	Chunk 1	Chunk 2	GPT-4o Mini	GPT-4o	8B	70B	8B	70B	Importance

Table S6. Effect sizes of features in LLM writing
Paired Cohen's *d* relative to human chunk 2

	GPT		Llama 3 Instruct		Llama 3 Base		
Feature	GPT-4o Mini	GPT-4o	8B	70B	8B	70B	Importance
Present participial clauses	1.37	1.38	0.44	0.56	-0.03	0.01	1,715.1
'That' clauses as subject	1.03	0.77	0.35	0.33	-0.22	-0.20	1,387.2
Mean word length	1.98	2.24	0.14	0.45	-0.19	0.01	1,070.4
Adverbs	-0.43	-0.53	-0.77	-0.77	0.04	0.05	916.0
Nominalizations	1.18	1.23	0.49	0.59	-0.15	-0.12	808.9
Phrasal co-ordination	0.44	0.81	0.61	0.53	-0.05	-0.02	776.0
Clausal co-ordination	-0.54	-0.59	0.47	0.36	0.21	0.18	727.3
Other nouns	-0.10	0.10	-0.38	-0.24	-0.09	-0.18	725.5
Downtoners	0.33	0.11	-0.26	-0.29	-0.19	-0.18	712.8
Attributive adjectives	0.91	1.09	-0.01	0.10	-0.41	-0.37	677.7
Demonstratives	0.39	0.32	-0.24	-0.21	-0.25	-0.21	675.9
Agentless passives	-0.59	-0.56	-0.04	-0.13	0.01	-0.02	664.4
Past participial postnominal	0.65	0.55	-0.13	-0.13	0.13	0.14	633.7
Infinitives	-0.24	-0.30	0.51	0.47	0.22	0.16	614.9
Prepositional phrases	0.77	0.68	-0.01	0.02	-0.46	-0.09	581.2
Emphatics	-0.31	-0.39	-0.03	-0.02	-0.23	-0.26	545.9
Existential 'there'	-0.27	-0.19	-0.37	-0.39	0.03	0.05	527.3
Possibility modals	0.38	0.04	0.08	0.13	-0.01	-0.01	521.1
Past tense	-0.41	-0.29	-0.12	-0.27	0.19	0.16	519.0
'That' verb complements	-0.23	-0.35	0.22	0.08	0.26	0.22	503.4
Contractions	-0.49	-0.50	0.35	0.38	0.29	0.24	497.7
Other adverbial subordinators	0.41	0.21	0.13	0.07	-0.18	-0.11	496.6
'Be' as main verb	-0.94	-0.83	0.11	0.00	0.09	0.02	494.3
Public verbs	-0.48	-0.37	-0.34	-0.34	0.13	0.10	466.1
Present participial postnominal	0.65	0.52	0.10	0.10	0.01	0.06	451.0
Predicative adjectives	-0.18	-0.11	0.32	0.28	0.24	0.23	450.8
Suasive verbs	-0.02	0.12	-0.07	-0.07	0.16	0.10	449.6
Analytic negation	-0.35	-0.51	-0.23	-0.27	0.09	0.05	439.6
Third-person pronouns	-0.14	-0.13	0.10	0.05	0.24	0.23	436.8
Private verbs	-0.17	-0.26	0.29	0.20	0.28	0.27	435.9
Perfect aspect	-0.42	-0.43	0.16	0.09	-0.03	-0.07	434.2
Pro-verb 'do'	-0.63	-0.62	-0.30	-0.33	0.06	0.04	430.8
Present tense	-0.40	-0.50	-0.01	0.00	0.28	0.27	418.9
Amplifiers	-0.25	-0.38	-0.33	-0.38	-0.07	-0.09	412.6
Pronoun 'it'	-0.17	-0.11	0.06	0.06	0.06	0.04	412.0
Gerunds	0.38	0.35	0.11	0.15	-0.09	-0.09	402.9
First-person pronouns	-0.32	-0.62	0.15	0.12	0.35	0.30	393.4
Because	-0.50	-0.50	-0.34	-0.35	0.08	0.02	387.9
Split auxiliaries	-0.08	-0.20	0.17	0.13	-0.01	-0.06	379.4
Time adverbials	-0.21	-0.19	-0.26	-0.26	0.07	0.06	366.0
Though, although	-0.13	0.09	-0.32	-0.28	-0.05	-0.03	361.2
Place adverbials	0.35	0.33	0.01	-0.01	0.11	0.17	353.9
Demonstrative pronouns	-0.50	-0.56	-0.28	-0.26	-0.01	-0.03	350.8
Predictive modals	-0.28	-0.43	-0.03	0.04	0.11	0.08	345.4
Indefinite pronouns	-0.22	-0.18	-0.11	-0.12	0.19	0.20	323.5
by-passives	-0.21	-0.17	0.06	0.03	0.04	0.01	314.9
Synthetic negation	-0.37	-0.27	-0.33	-0.34	-0.02	-0.04	311.8
'Wh-' questions	-0.06	-0.25	0.16	0.16	0.11	0.10	306.1

Table S6. Effect sizes of features in LLM writing (cont'd)

	GPT		Llama 3 Instruct		Llama 3 Base		
Feature	GPT-4o Mini	GPT-4o	8B	70B	8B	70B	Importance
If, unless	-0.16	-0.30	-0.29	-0.27	0.09	0.06	305.1
'Seem' and 'appear'	0.12	0.27	0.12	0.09	0.00	0.02	290.3
Necessity modals	0.02	-0.10	-0.22	-0.22	0.14	0.12	286.7
Conjuncts	0.07	-0.03	-0.02	0.09	-0.01	0.02	281.2
Second-person pronouns	-0.39	-0.49	-0.21	-0.20	0.11	0.07	277.3
Stranded prepositions	-0.16	-0.16	0.04	0.05	-0.04	-0.07	266.5
'Wh-' relatives as subject	-0.21	-0.23	0.04	0.01	0.03	0.01	259.1
Sentence relatives	-0.26	-0.25	0.02	0.06	0.00	-0.03	251.7
'Wh-' clauses	-0.17	-0.24	-0.06	-0.14	0.11	0.10	247.1
Discourse particles	-0.20	-0.20	-0.04	-0.04	0.14	0.04	181.9
Hedges	-0.22	-0.16	-0.15	-0.13	-0.04	-0.02	181.1
Past participial clauses	0.33	0.36	0.11	0.10	-0.09	-0.07	171.7
'That' deletion	-0.11	-0.16	-0.10	-0.11	0.09	0.11	169.8
Pied-piping relative clauses	-0.16	-0.17	-0.23	-0.21	-0.17	-0.17	164.5
'That' clauses as object	-0.13	-0.17	-0.12	-0.15	-0.02	-0.03	118.3
'That' adjective complements	-0.06	-0.16	0.05	0.02	0.09	0.06	107.6
Split infinitives	-0.05	-0.02	0.02	0.01	-0.15	-0.13	94.9
'Wh-' relatives as object	-0.25	-0.27	-0.28	-0.26	-0.10	-0.10	62.5

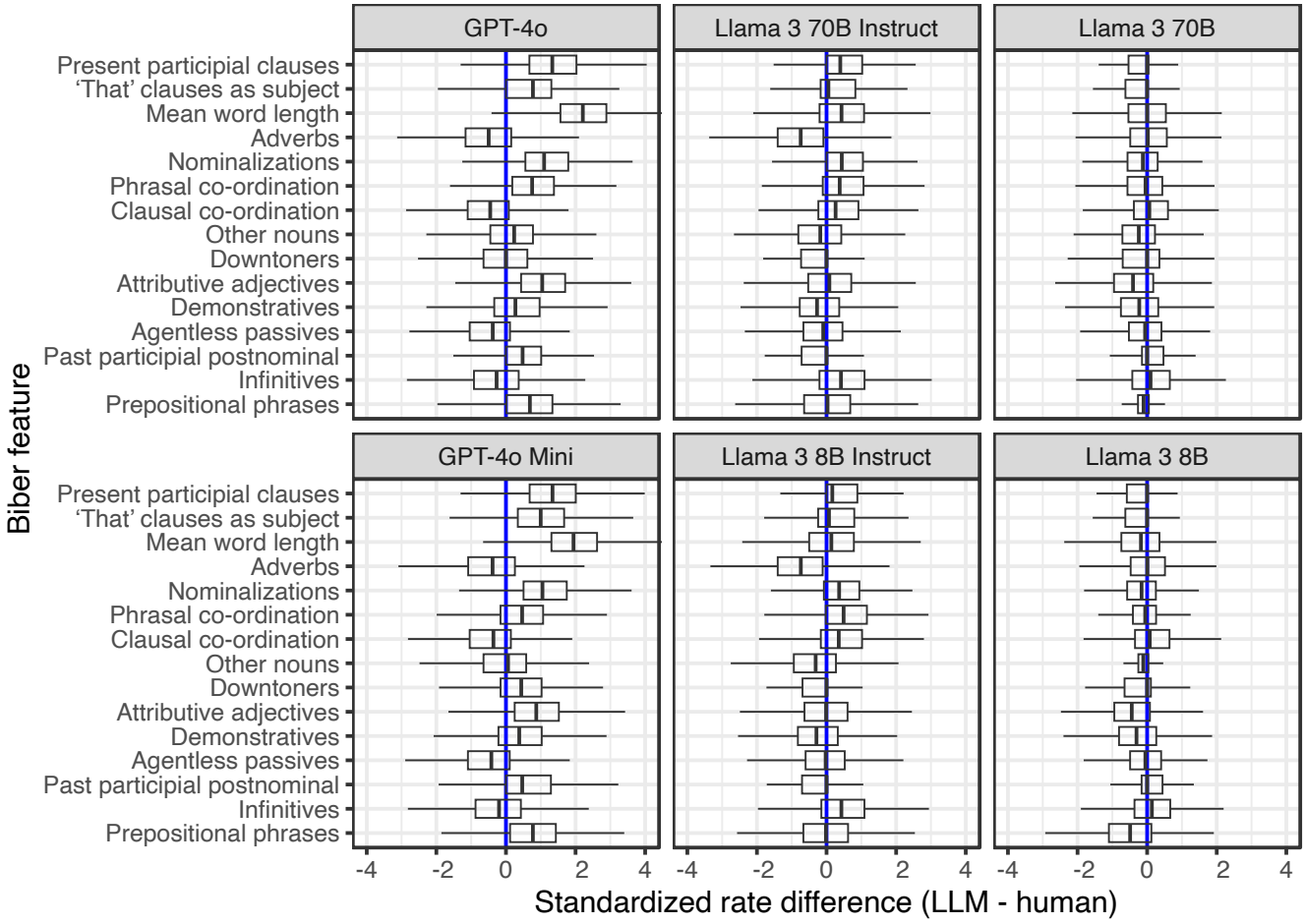


Fig. S1. Boxplots of paired differences between features for LLM and human writing following the same chunk 1, standardized by the standard deviation of the paired differences. Some features show very little overlap: for instance, with GPT-4o and GPT-4o Mini, the mean word length of the LLM-generated text is almost always longer than the mean word length in the corresponding human chunk 2.

Table S7. Most overrepresented words in LLM-generated texts (by frequency), ordered by fraction of outputs containing each word

GPT-4o		GPT-4o Mini		Llama 3 70B Instruct		Llama 3 8B Instruct		Llama 3 70B		Llama 3 8B	
Word	Rate	Word	Rate	Word	Rate	Word	Rate	Word	Rate	Word	Rate
amidst	27%	amidst	27%	reminder	12%	reminder	15%	bananas	8%	deborah	4%
tapestry	23%	tapestry	24%	prioritize	5%	unease	6%	bam	6%	goodnight	3%
intricate	14%	vibrant	20%	pang	4%	prioritize	5%	denis	3%	matty	3%
underscore	14%	ignite	20%	palpable	4%	pang	4%	darry	2%	ml	2%
unspoken	11%	intricate	17%	unease	4%	continuation	4%	paperback	2%	rambo	2%
camaraderie	11%	palpable	15%	policymaker	3%	palpable	4%	verona	2%	merlin	1%
solace	10%	fleeting	15%	intricate	3%	waft	2%	rekall	2%	missy	1%
unravel	10%	camaraderie	13%	continuation	2%	rut	2%	filth	1%	elisa	1%
fleeting	9%	grapple	12%	shoutout	2%	grapple	2%	janice	1%	fay	1%
palpable	9%	cacophony	7%	camaraderie	1%	camaraderie	1%	ebook	1%	worcester	1%

Table S8. Most underrepresented words in LLM-generated texts, relative to human usage rates

GPT-4o		GPT-4o Mini		Llama 3 70B Instruct		Llama 3 8B Instruct		Llama 3 70B		Llama 3 8B	
Word	Rate	Word	Rate	Word	Rate	Word	Rate	Word	Rate	Word	Rate
i.e.	0.0026	extremely	0.0034	yep	0.0075	anyways	0.0097	jeez	0.040	bingo	0.033
blah	0.0063	spokesman	0.0059	fucking	0.0075	analyse	0.011	donnie	0.042	scorch	0.040
fuck	0.0079	i.e.	0.0095	horrible	0.0077	somebody	0.012	alexithymia	0.046	been	0.043
fucking	0.0083	unhappy	0.010	ok	0.011	characterise	0.013	frampton	0.048	abt	0.043
asshole	0.0093	bitch	0.012	fuckin	0.013	fuck	0.014	y'all	0.048	bananas	0.043
nasty	0.011	fucking	0.012	amid	0.015	obviously	0.014	analogous	0.053	pham	0.043
chum	0.013	kg	0.013	fortunately	0.017	ok	0.014	carotenoid	0.053	ie	0.045
shit	0.014	fuck	0.014	i.e.	0.018	fuckin	0.014	mussel	0.053	monstrous	0.045
ok	0.014	visa	0.015	obviously	0.018	blah	0.015	que	0.053	pow	0.045
and/or	0.016	ter	0.016	amongst	0.019	yep	0.016	whatsapp	0.053	unquote	0.045

Table S9. Random forest pairwise classification accuracy, distinguishing each LLM from human text. Training accuracy based on out-of-bag error

	Trained on HAP-E		Trained on CAP	
LLM	Training	Test on CAP	Training	Test on HAP-E
GPT-4o				
GPT-4o	98.3%	98.4%	98.9%	97.4%
GPT-4o Mini	98.7%	98.2%	99.2%	97.5%
Llama Instruct				
Llama 3 70B Instruct	94.5%	93.5%	94.6%	91.8%
Llama 3 8B Instruct	95.5%	95.3%	96.1%	94.1%
Llama Base				
Llama 3 70B	93.0%	94.6%	95.8%	89.1%
Llama 3 8B	94.0%	93.6%	95.6%	89.4%

Table S10. Lasso pairwise classification accuracy, distinguishing each LLM from human text

	Trained on CAP		Trained on HAP-E	
LLM	Training	Test on HAP-E	Training	Test on CAP
GPT-4o				
GPT-4o	95.9%	95.3%	96.5%	95.3%
GPT-4o Mini	96.7%	96.3%	96.9%	96.2%
Llama Instruct				
Llama 3 70B Instruct	91.1%	90.7%	92.1%	90.2%
Llama 3 8B Instruct	92.4%	91.8%	92.9%	91.9%
Llama Base				
Llama 3 70B	77.2%	73.2%	75.6%	75.3%
Llama 3 8B	77.6%	73.8%	76.3%	75.3%

Table S11. Random forest accuracy when trained on HAP-E and classifying M4 arXiv data.

Model	Acc. (in-sample)	Acc. (out-of-sample)
ChatGPT		
GPT 4o	98.22%	70.63%
GPT 4o Mini	98.73%	60.77%
Llama Instruction-Tuned		
Llama 3 70B Instruct	95.33%	68.20%
Llama 3 8B Instruct	96.20%	65.60%
Llama Base		
Llama 3 70B	92.43%	51.17%
Llama 3 8B	93.56%	51.85%

Table S12. Random forest accuracy when trained on CAP and classifying M4 arXiv data.

Model	Acc. (in-sample)	Acc. (out-of-sample)
ChatGPT		
GPT 4o	98.76%	70.68%
GPT 4o Mini	99.04%	65.27%
Llama Instruction-Tuned		
Llama 3 70B Instruct	95.35%	59.27%
Llama 3 8B Instruct	96.60%	57.95%
Llama Base		
Llama 3 70B	95.53%	50.92%
Llama 3 8B	95.39%	51.48%

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